

Individual Project
Data Mining B141

GitHub repository: <https://github.com/danielarubio2005/b141-data-mining>

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1. Project Objective and Scope

1.1 Business Context

This project is conducted from the perspective of a newly appointed data scientist at a UK-based online gift retailer that sells a wide range of decorative homeware and gift products to both individual consumers and wholesale buyers across Europe (Chen, 2019). Management has requested a data-driven analysis of the company’s transaction history to identify non-trivial patterns that can be used to improve products, increase customer retention, and boost sales.

Hence, the primary deliverable of this analysis is a set of actionable insights, communicated in terms accessible to non-technical stakeholders, and backed by a fully documented and reproducible implementation and GitHub Repository.

1.2 Dataset Description

The dataset selected for this project is the UCI Online Retail II dataset, a publicly available real-world transactional dataset donated in 2019 and published by the UCI Machine Learning Repository (Chen, 2019). It contains invoice-level records from an online retail business operating between 1 December 2009 and 9 December 2011.

Source: <https://archive.ics.uci.edu/dataset/502/online+retail+ii>

Key statistics: 1,067,371 raw transaction records, 8 features, File format: .xlsx (two sheets by year).

Column	Type	Description
Invoice	String	Unique transaction ID. Prefix “C” indicates a cancellation.
StockCode	String	Unique product/item code.
Description	String	Human-readable product name (e.g. “WHITE HANGING HEART T-LIGHT HOLDER”).
Quantity	Integer	Number of units purchased per transaction line. Negative values indicate returns.
InvoiceDate	DateTime	Date and time the transaction was recorded.
Price	Float	Unit price in GBP (£).
Customer ID	Float	Unique customer identifier. Missing for ~25% of records (guest checkouts).
Country	String	Country of the customer. Predominantly United Kingdom.

Table 1: Data set feature overview.

As seen in Figure 1, when loading the dataset.

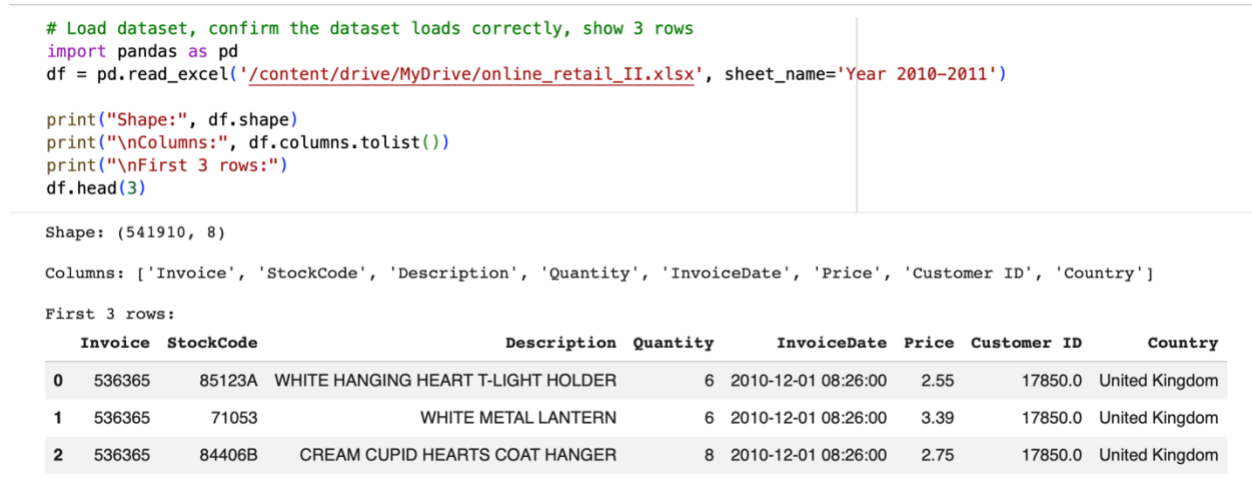


Figure 1: Screenshot of cell 1 of project's notebook. Load dataset, confirms the loading is correct and print first 3 rows. Shows figures of the dataset.

For this project, analysis focuses on the 2010–2011 sheet, which contains approximately 541,000 records and is used as the primary working dataset. This provides a sufficiently large sample to yield meaningful patterns.

1.3 Research Questions and Scope

The project addresses three specific research questions, each mapped to a relevant data mining technique:

Research Question	Technique	Business Value
Which products are most frequently purchased together?	Association Rule Mining (Apriori)	Enables product bundling, cross-selling recommendations, and shelf placement decisions.
Are there abnormal transactions in the data?	Outlier Detection (Z-score)	Identifies data quality issues, potential fraud, or unusual bulk orders requiring review.
Can customers be grouped by purchasing behaviour?	RFM Analysis + K-Means Clustering	Supports targeted marketing campaigns and personalized retention strategies.

Table 2: Research questions and corresponding data mining techniques.

Together, these three questions cover the breadth of the data mining pipeline: pattern discovery (association rules), anomaly detection (outlier analysis), and predictive segmentation (clustering).

1.4 Scope and Limitations

The following boundaries define the scope of this project:

- Only the 2010–2011 data sheet is used. Results may differ if the 2009–2010 sheet is incorporated, as product lines and pricing may have changed.
- Customer-level analysis (RFM clustering) excludes the approximately 25% of transactions where no Customer ID is recorded, as these cannot be attributed to a specific individual.
- No external data sources (e.g. economic indicators, competitor data, or marketing spend) are incorporated. All findings are derived solely from the transaction log.
- This project does not build a real-time or productionised system. All analysis is exploratory and retrospective, intended to surface insights rather than automate decisions.

2. Data Collection and Preprocessing

2.1 Overview

Before any mining technique can be applied, the raw data must be audited, cleaned, transformed, and enriched. Therefore, this section documents every preprocessing decision made, with the rationale grounded in data quality principles; the preprocessing pipeline consists of six sequential steps:

- Step 1: Data loading and initial inspection
- Step 2: Removing missing values
- Step 3: Removing cancelled and invalid transactions
- Step 4: Type correction
- Step 5: Feature engineering (attribute construction)
- Step 6: Normalization for clustering

2.2 Step 1: Data Loading and Inspection

The dataset is loaded from the local Google Drive mount using the pandas library. Only the 2010–2011 sheet is used, as defined in the project scope. Then, an initial inspection is performed to understand the structure, data types, and extent of missing values before any transformations are applied.

```
# A copy of df from the previous cell that now becomes df_raw
# This protects the original so I can always go back to it
df_raw = df.copy()

print(f"Raw shape: {df_raw.shape[0]:,} rows × {df_raw.shape[1]} columns")
print("\nMissing values:")
print(df_raw.isnull().sum())
print("\nNumeric summary:")
print(df_raw.dtypes)
df_raw[['Quantity', 'Price']].describe().round(2)
```

*** Raw shape: 541,910 rows × 8 columns

Missing values:

Invoice	0
StockCode	0
Description	1454
Quantity	0
InvoiceDate	0
Price	0
Customer ID	135080
Country	0

dtype: int64

Numeric summary:

Invoice	object
StockCode	object
Description	object
Quantity	int64
InvoiceDate	datetime64[ns]
Price	float64
Customer ID	float64
Country	object

dtype: object

	Quantity	Price
count	541910.00	541910.00
mean	9.55	4.61
std	218.08	96.76
min	-80995.00	-11062.06
25%	1.00	1.25
50%	3.00	2.08
75%	10.00	4.13
max	80995.00	38970.00

Figure 2: Initial data quality inspection for pre-processing. Screenshot of cell 4 of project's notebook.

After running this cell, the following initial data quality issues are revealed:

- Customer ID: 135.080 missing values (~25% of all rows). These correspond to guest or unregistered checkouts.
- Description: 1.454 missing values. Product names are occasionally absent.
- Quantity and Price: Both contain negative values. Negative quantities represent returned goods, and negative prices could be erroneous test records.
- CustomerID: Stored as a float type rather than a string, and it is a categorical identifier not a numeric value.

2.3 Step 2: Handling Missing Values

To handle missing values is possible to input them (fill them with estimates such as the mean or mode) or drop rows where imputation is not meaningful. In this case, imputation is not appropriate because missing Customer ID values cannot be inferred, as they represent anonymous transactions for which no customer identity exists, so these rows are excluded from all customer-level analyses (RFM clustering) to avoid introducing false associations, and missing description

values affect fewer than 1% of rows and correspond to non-product entries (e.g. postage charges or manual adjustments), hence these are dropped (as shown in Figure 3).

```
df = df_raw.copy()

# Remove rows with no Customer ID (guest checkouts - can't identify who bought)
df = df.dropna(subset=['Customer ID'])

# Remove rows with no Description (non-product entries like postage)
df = df.dropna(subset=['Description'])

print(f"After removing missing values: {len(df):,} rows")
```

After removing missing values: 406,830 rows

Figure 3: Handling missing values. Screenshot of cell 5 of project's notebook.

2.4 Step 3: Removing Cancelled and Invalid Transactions

The dataset contains two further categories of invalid records that must be removed before analysis:

- **Cancelled Orders:** Invoices prefixed with the letter 'C' (e.g. 'C536379') represent cancellations of previously placed orders, but including these records would distort both the association rule mining (by appearing to co-purchase products that were actually returned) and the customer revenue calculations used in RFM analysis.
- **Invalid Quantities and Prices:** It is not possible to have rows with non-positive quantities or removed prices. Indeed, a Quantity of zero or below implies a return, error, or correction entry, while a Price of zero or below is inconsistent with a genuine retail sale.

Thus, as shown in Figure 4, both were removed.

```
# Cancelled orders have Invoice starting with 'C' (e.g. C536379)
df = df[~df['Invoice'].astype(str).str.startswith('C')]

# Remove returns (negative quantity) and bad price entries (zero or negative)
df = df[(df['Quantity'] > 0) & (df['Price'] > 0)]

print(f"After removing cancellations and invalid records: {len(df):,} rows")
```

After removing cancellations and invalid records: 397,885 rows

Figure 4: Removing canceled Orders and invalid Quantities and Prices. Screenshot of cell 6 of project's notebook.

2.5 Step 4: Type Correction

The CustomerID column requires type correction to enable downstream analysis. As noted before, Customer ID is stored as a float (e.g. 17850.0), but it should be a string as it is an identifier, so it is cast to integer and then to string, as shown in Figure 5.

```
# Customer ID is a float (17850.0) so I convert to string since it's an ID not a number
df['Customer ID'] = df['Customer ID'].astype(int).astype(str)

print(df.dtypes)
```

...	Invoice	object
	StockCode	object
	Description	object
	Quantity	int64
	InvoiceDate	datetime64[ns]
	Price	float64
	Customer ID	object
	Country	object
	dtype:	object

Figure 5: Customer ID type correction. Screenshot of Cell 7 of project's notebook.

2.6 Step 5: Feature Engineering

The new attributes that capture important information more effectively than the original ones were created, so two derived features are added:

- Revenue: The product of Quantity and Price, giving the monetary value of each transaction line. This is the primary numeric target for EDA and RFM analysis.
- Month: A Period column derived from InvoiceDate, used to aggregate and visualize monthly sales trends in the EDA section.

```
# Revenue = how much money each line item made
df['Revenue'] = df['Quantity'] * df['Price']

# Month = used later to plot sales trends over time
df['Month'] = df['InvoiceDate'].dt.to_period('M')

print(df[['Quantity', 'Price', 'Revenue', 'Month']].head(5))
print(f"\nTotal revenue in dataset: £{df['Revenue'].sum():,.2f}")
```

	Quantity	Price	Revenue	Month
0	6	2.55	15.30	2010-12
1	6	3.39	20.34	2010-12
2	8	2.75	22.00	2010-12
3	6	3.39	20.34	2010-12
4	6	3.39	20.34	2010-12

Total revenue in dataset: £8,911,425.90

Figure 6: Derived features added, Revenue and Month, for upcoming analysis. Screenshot fo Cell 8 of project's notebook.

2.7 Step 6: Normalization for Clustering

For the customer segmentation task (Section 4.3), the RFM features (Recency, Frequency, Monetary) operate on very different scales: Recency is measured in days (range roughly 1-365), Frequency in number of invoices (range 1-200+), and Monetary in pounds sterling (range £1-£250,000+). Therefore, Z-score normalization (StandardScaler) is applied so that no single feature dominates the K-Means distance calculation simply because of its scale; each feature is transformed to have zero mean and unit standard deviation: $v' = (v - \mu) / \sigma$.

This step is applied at the point of clustering in Section 4 rather than to the full dataset, to preserve human-interpretable values throughout the EDA and association rule steps, however it is showed in Figure 7.

```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

# Normalise so no single feature dominates by scale
scaler = StandardScaler()
rfm_scaled = scaler.fit_transform(rfm[['Recency', 'Frequency', 'Monetary']])

# Run KMeans with 4 clusters
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)

# Show average RFM values per cluster
print(rfm.groupby('Cluster')[['Recency', 'Frequency', 'Monetary']].mean().round(1))
```

	Recency	Frequency	Monetary
Cluster			
0	43.7	3.7	1359.1
1	248.1	1.6	480.6
2	7.4	82.5	127338.3
3	15.5	22.3	12709.1

Figure 7: Normalisation for Clustering. Screenshot of Cell 25 of project's notebook.

2.8 Preprocessing Summary

Table 3 below summarizes the impact of each cleaning step on the dataset size, providing a transparent record of data reduction decisions.

Metric	Before Cleaning	After Cleaning
Raw dataset loaded (2010–11 sheet)	541,910	-
After removing missing Customer ID & Description	406,830	-24.93%
After removing cancelled orders (Invoice = C*) & Quantity & Price (≤ 0)	397,885	-2.20%
Final clean dataset	397,885	73.42% of raw

Table 3: Preprocessing timeline: Row count at each stage.

The final clean dataset of 397,885 records retains 73.42% of the raw 2010–11 sheet. The largest single reduction comes from the 135,080 guest-checkout rows where no Customer ID was recorded, considering these cannot be attributed to an individual and are therefore excluded from customer-level analysis. Then, the remaining removals (cancellations, invalid quantities and prices) account for a further 2.2% of records. To sum up, the resulting dataset contains zero missing values and is fully suitable for the mining techniques applied in Sections 3 and 4, as shown in Figure 8.

```
print("=" * 45)
print("CLEAN DATASET SUMMARY")
print("=" * 45)
print(f"Rows: {len(df):,}")
print(f"Missing values: {df.isnull().sum().sum()}")
print(f"Unique customers: {df['Customer ID'].nunique():,}")
print(f"Unique products: {df['StockCode'].nunique():,}")
print(f"Unique invoices: {df['Invoice'].nunique():,}")
print(f"Countries: {df['Country'].nunique():,}")
print(f>Date range: {df['InvoiceDate'].min().date()} → {df['InvoiceDate'].max().date()}")
print(f"Total revenue: £{df['Revenue'].sum():,.2f}")

=====
CLEAN DATASET SUMMARY
=====
Rows:      397,885
Missing values: 0
Unique customers: 4,338
Unique products: 3,665
Unique invoices: 18,532
Countries: 37
Date range: 2010-12-01 → 2011-12-09
Total revenue: £8,911,425.90
```

Figure 8: Key statistics of cleaned data set. Screenshot of cell 9 of project's notebook.

3. Exploratory Data Analysis

3.1 Overview

Exploratory Data Analysis (EDA) is the process of examining a dataset through statistical summaries and visualisations to identify patterns, trends, and irregularities before applying formal mining techniques (IBM, n.d.). Accordingly, data visualisation maps data onto graphical primitives to provide a qualitative overview, search for patterns and relationships, and identify suitable parameters for further analysis. Therefore, six analyses are performed below, considering the dataset and research questions that the project aim to analyze.

3.2 Statistical Summary

First, a basic statistical review was performed; Table 4 presents basic statistical descriptions of the three key numeric columns after cleaning.

	Quantity	Price	Revenue
count	397885.00	397885.00	397885.00
mean	12.99	3.12	22.40
std	179.33	22.10	309.07
min	1.00	0.00	0.00
25%	2.00	1.25	4.68
50%	6.00	1.95	11.80
75%	12.00	3.75	19.80
max	80995.00	8142.75	168469.60

Table 4: Statistical descriptions of the three key numeric columns. Screenshot of cell 12 of project's notebook.

The mean Revenue (£22.40) is notably higher than the median (£11.80), indicating a right-skewed distribution, meaning that a small number of high-value line items pull the mean upward. This is confirmed later by Figure 11.

3.3 Top Products by Revenue

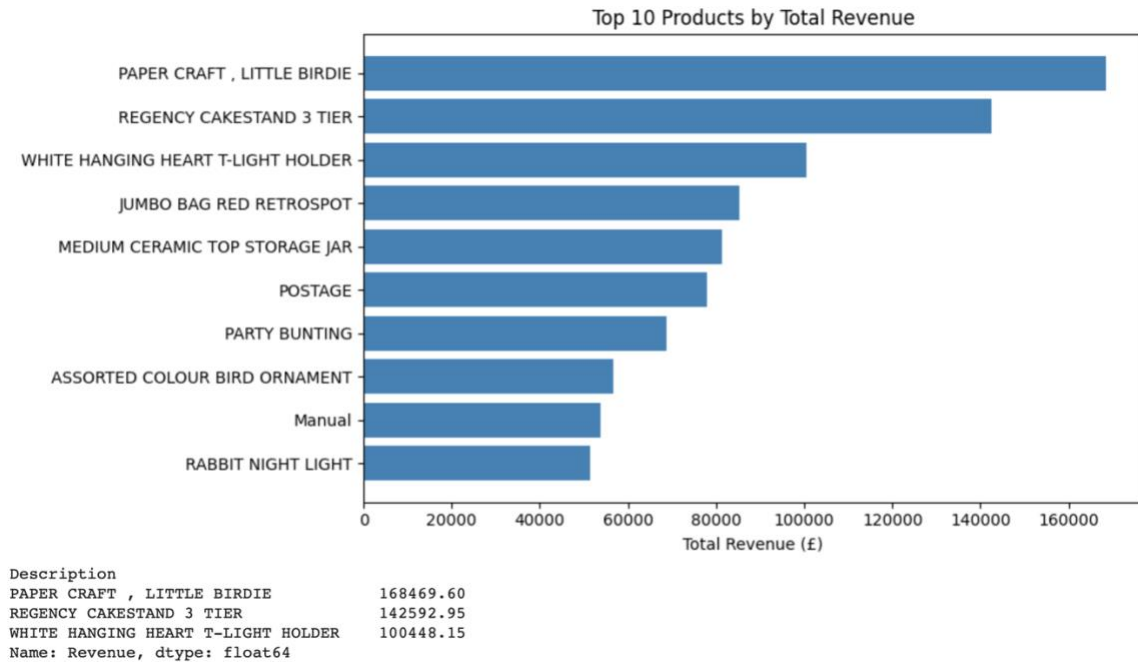


Figure 9: Top 10 Products by Total Revenue. Screenshot of cell 13 of project's notebook.

Figure 8 shows the ten products generating the highest cumulative revenue. Notably, “PAPER CRAFT, LITTLE BIRDIE” accounts for £168469.60 in total revenue, making it the single largest revenue driver in the dataset. Moreover, the top 10 products are dominated by decorative homeware items, consistent with the retailer's product focus. This finding directly informs the association rule analysis in Section 4, where these high-frequency products are expected to appear as strong rule antecedents.

3.4 Revenue by Country

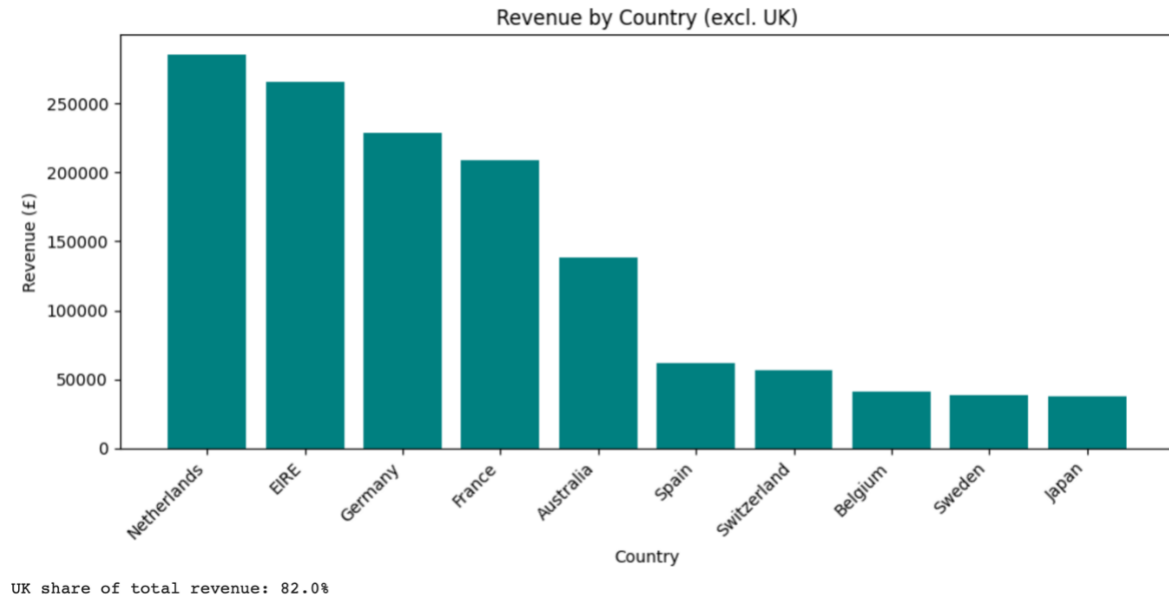


Figure 10: Revenue by Country (excl. UK). Screenshot of cell 14 of project's notebook.

(UK is excluded from the graph because it is dominant, so the other countries are visible.)

The United Kingdom accounts for 82% of total revenue, reflecting the retailer's domestic customer base, however, among international markets, Netherlands and EIRE (Ireland) are the largest contributors. Indeed, the dominance of the UK market means the association rules derived in Section 4 will primarily reflect British consumer purchasing behavior.

3.5 Monthly Revenue Trend

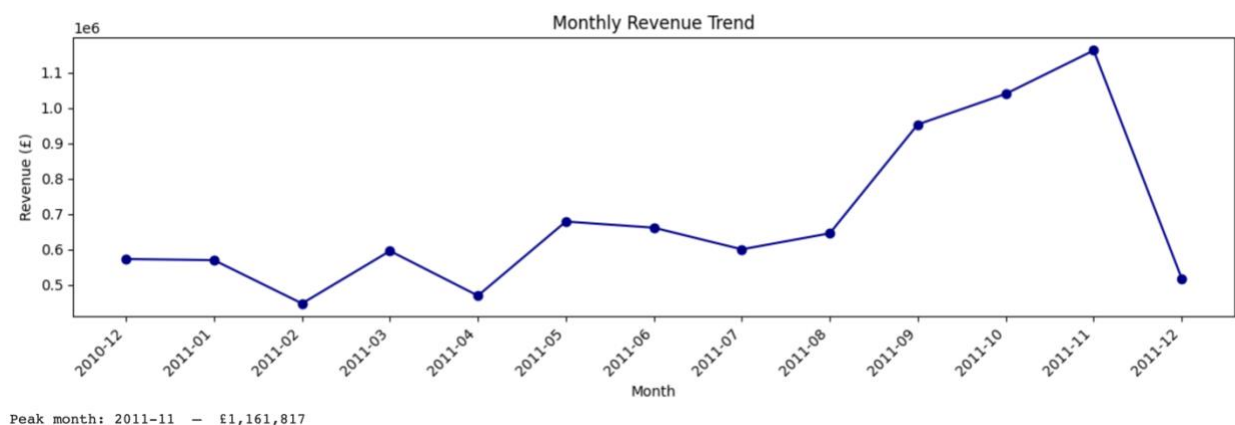


Figure 11: Monthly Revenue Trend. Screenshot of cell 15 of project's notebook.

Figure 10 reveals a clear seasonal pattern in revenue; sales remain relatively stable from December 2010 through August 2011, before rising sharply in September-November 2011,

peaking at £1,161,817 in November. Nevertheless, this pattern is consistent with pre-Christmas purchasing behavior typical of a gift retailer. Furthermore, it's important to note that the sharp drop in December 2011 likely reflects incomplete data for that month rather than an actual sales decline, which then its also a limitation for the project.

3.6 Revenue Distribution

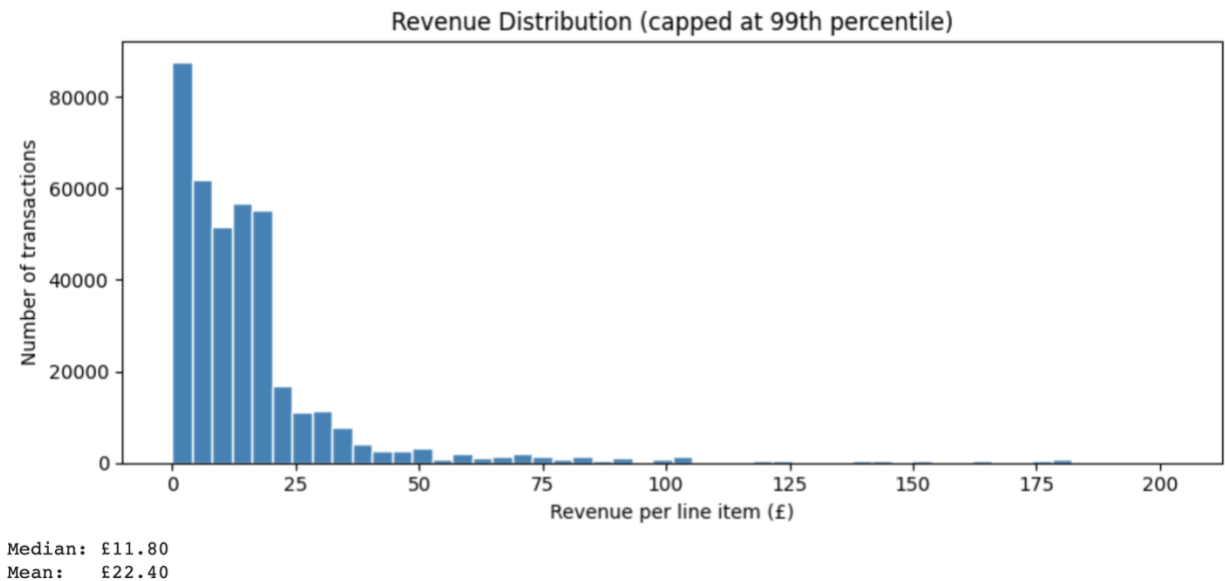


Figure 12: Revenue Distribution (Capped at 99th percentile). Screenshot of cell 16 of project's notebook.

(Cap at 99th percentile so extreme outliers don't squash the chart.)

The revenue distribution is strongly right-skewed (as noted before): most line items generate under £25, while a small number of bulk or high-value orders generate significantly more. This skew has practical implications; the outlier detection technique applied in Section 4.2 specifically targets these extreme values.

3.7 Orders by Day of Week

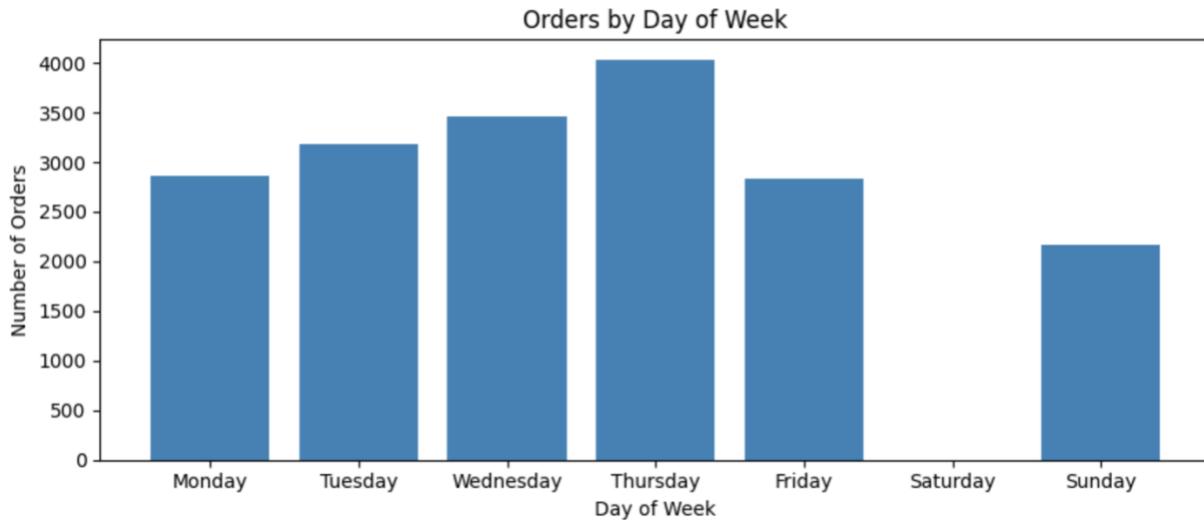


Figure 13: Orders by Day of the Week. Screenshot of cell 17 of project's notebook.

As shown in Figure 12, Thursday records the highest order volume (~4,000 orders), followed by Wednesday and Tuesday. Saturday records virtually no orders, indicating the business does not operate on weekends, however Sunday shows a small but notable volume (~2,150 orders), which may reflect online orders placed by customers outside business hours. This temporal insight could inform decisions about email marketing timing and promotional scheduling, for example, campaigns launched mid-week, particularly on Thursday, are likely to reach customers at peak engagement.

4. Data Mining Techniques

4.1 Association Rule Mining (Apriori)

To implement the Apriori algorithm, the dataset was filtered to UK transactions only, producing a basket matrix of 16,646 invoices and 3,844 products. Then, a minimum support threshold of 2% was applied, yielding 199 frequent items. Afterwards, candidate pairs were evaluated for support, confidence, and lift, hence a total of 35 rules met the minimum support threshold. Accordingly, the top rule by lift is: customers who purchased “GREEN REGENCY TEACUP AND SAUCER” also purchased “PINK REGENCY TEACUP AND SAUCER” with a support of 0.0243, confidence of 0.6601, and lift of 22.2891, meaning this co-purchase is 22.28 times more likely than random chance, as shown by Figure 13. These findings suggest strong bundling opportunities for the retailer, which will be further explored in the next section.

Association rules found: 35

	Item_A	Item_B	Support	Confidence	Lift
0	GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0243	0.6601	22.2891
1	ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.0286	0.7021	19.0957
2	ROSES REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.0230	0.5649	19.0736
3	GARDENERS KNEELING PAD KEEP CALM	GARDENERS KNEELING PAD CUP OF TEA	0.0275	0.6173	16.3872
4	ALARM CLOCK BAKELIKE RED	ALARM CLOCK BAKELIKE GREEN	0.0273	0.5989	14.4493
5	WOODEN PICTURE FRAME WHITE FINISH	WOODEN FRAME ANTIQUE WHITE	0.0275	0.5363	11.3868
6	PAPER CHAIN KIT 50'S CHRISTMAS	PAPER CHAIN KIT VINTAGE CHRISTMAS	0.0263	0.4624	11.3703
7	LUNCH BAG SPACEBOY DESIGN	LUNCH BAG WOODLAND	0.0208	0.3963	9.7884
8	HEART OF WICKER SMALL	HEART OF WICKER LARGE	0.0239	0.4289	9.0254
9	LUNCH BAG CARS BLUE	LUNCH BAG PINK POLKADOT	0.0238	0.4490	8.8133

Table 5: Results of Apriori algorithm application. Screenshot of cell 21 of project's notebook.

4.2 Outlier Detection

Z-score normalization was applied to total invoice revenue to identify anomalous transactions. Any invoice more than 3 standard deviations from the mean (threshold: £5,515.45) was flagged as an outlier, as a consequence, of 18,532 total invoices, 99 (0.53%) were identified as outliers, which seems a small amount yet these account for 14.5% of total revenue, representing a disproportionate share of revenue consistent with wholesale bulk purchasing behavior. Indeed, the largest outlier invoice totaled £168,469.60; these extreme transactions likely represent wholesale bulk orders rather than typical retail purchases, and their presence explains the large gap between mean (£480.87) and median (£303.04) invoice values observed in the EDA. Nevertheless, excluding them from customer segmentation analysis ensures the RFM clusters in Section 4.3 reflect genuine retail purchasing patterns rather than being distorted by small number of atypical orders.

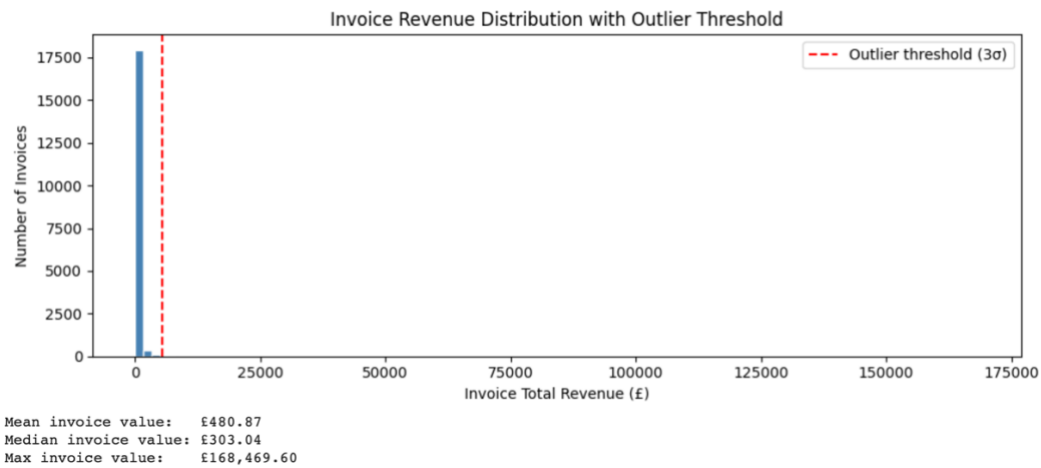


Figure 14: Invoice Revenue Distribution with Outlier Threshold. Screenshot of cell 24 of project's notebook.

4.3 Customer Segmentation

RFM analysis was applied to segment 4,338 customers into 4 groups using K-Means clustering. For this, features were normalized using z-score standardization (StandardScaler) prior to clustering to prevent scale dominance. Then, the resulting silhouette score of 0.616 indicates strong cluster separation.

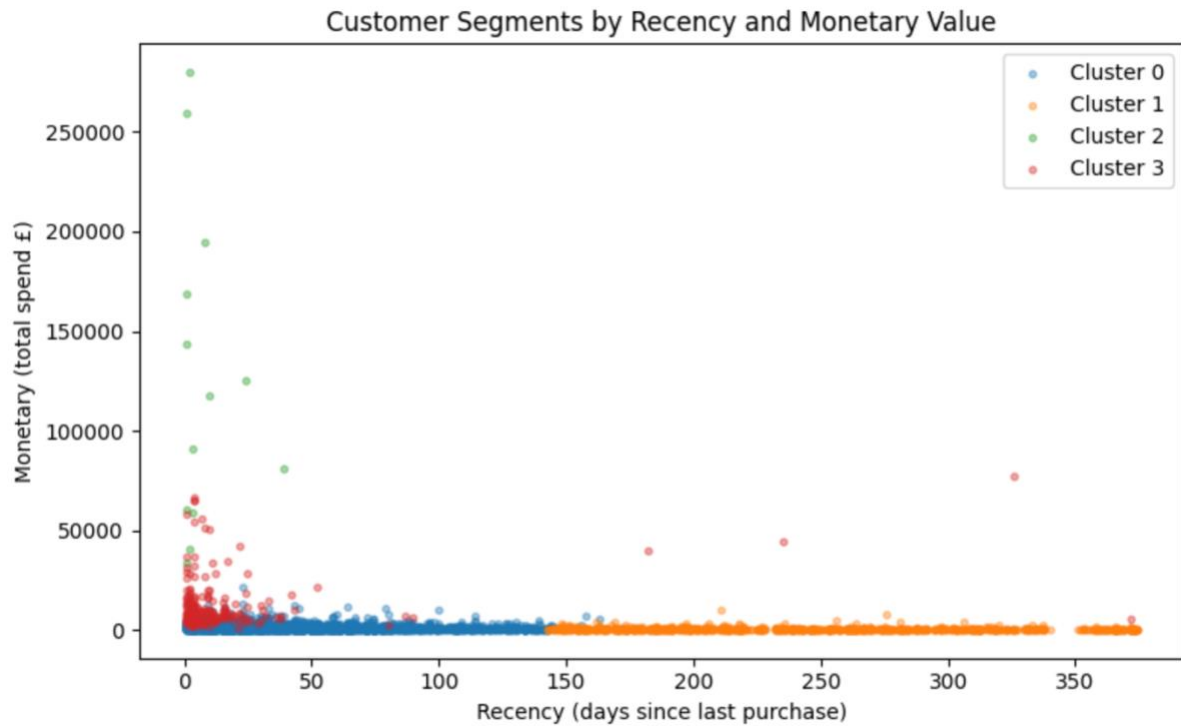


Figure 15: Customer Segments by Recency and Monetary Value. Screenshot of cell 28 of project's notebook.

Four distinct customer segments were identified: Champions (13 customers: purchased very recently, extremely high frequency and spend, likely wholesale accounts); Loyal Customers (204 customers: recent, high frequency, high spend, the retailer's core base); Potential Loyalists (3,054 customers: moderate recency, low frequency, low spend, representing the largest group with growth potential); and Lost Customers (1,067 customers: last purchased over 248 days ago on average, very low frequency and spend, at high risk of permanent churn), as shown in Table 6. These segments directly inform the business recommendations presented in Section 5.

Cluster	Recency	Frequency	Monetary	Size	Label
2	7.4 days	82.5 orders	£127,338	13	Champions
3	15.5 days	22.3 orders	£12,709	204	Loyal Customers
0	43.7 days	3.7 orders	£1,359	3,054	Potential Loyalists
1	248.1 days	1.6 orders	£481	1,067	Lost Customers

Table 6: Customer Segments (Clustering Results).

5. Evaluation and Recommendations

5.1 Evaluation of Techniques

5.1.1 Association Rule Mining

The Apriori implementation successfully identified 35 association rules from 199 frequent items at a 2% minimum support threshold. Moreover, the quality of the rules is assessed using three metrics: Support measures how frequently a pair co-occurs across all transactions; the top rule, “GREEN” and “PINK REGENCY TEACUP AND SAUCER”, achieves a support of 0.0243, meaning it appears in 2.4% of all UK invoices. Then, confidence of 0.6601 indicates that 66.01% of customers who purchased the green teacup also purchased the pink variant in the same transaction. Most critically, the lift of 22.2891 confirms this is not coincidental; this co-purchase is 22.28 times more likely than if the two products were bought independently. In fact, a lift value this far above 1.0 represents a very strong association and provides reliable grounds for a product bundling recommendation, suggesting that the technique was successfully applied and relevant for the dataset. However, a limitation of the Apriori algorithm is its computational cost; evaluating all candidate pairs grows exponentially as the number of items increases, which is why the basket was filtered to UK transactions only and a minimum support threshold was applied to reduce the search space to a manageable size.

5.1.2 Outlier Detection

The Z-score method identified 99 outlier invoices (0.53% of all transactions) that collectively account for 14.5% of total revenue. This disproportionate revenue concentration validates the decision to use a statistical threshold rather than manual inspection, since no human reviewer would reliably identify these 99 invoices from 18,532 without a systematic approach. The threshold of £5,515.45 (3 standard deviations above the mean) is a well-established convention in industry practice, however a limitation of this approach is that it assumes revenue follows a roughly normal distribution, which the EDA showed is not strictly the case given the right skew, therefore a more robust alternative would be IQR-based detection, which is less sensitive to extreme values.

5.1.3 Customer Segmentation

The K-Means clustering achieved a silhouette score of 0.616, which falls in the "strong" range (above 0.5), indicating that customers within each cluster are meaningfully more similar to each other than to members of other clusters. Additionally, the four segments are well-separated and interpretable, particularly the Champions cluster (13 customers, average spend £127,338) and the Lost Customers cluster (1,067 customers, average recency 248 days). Nonetheless, a limitation is that K-Means requires the number of clusters to be specified in advance (4 was chosen based on

RFM convention), but an elbow method analysis could be used in future work to determine the optimal number empirically.

5.2 Business Recommendations

Based on the three mining techniques applied, four actionable recommendations are proposed for the retailer:

- **Product Bundling:** The association rules reveal strong co-purchase patterns within the Regency teacup and saucer range (lift = 22.28), thus the retailer should create bundled product listings and gift sets combining these items, and configure the website to display "frequently bought together" suggestions for customers viewing either product individually, as well as for the other variations of cups of tea and saucers as they also are often purchased together according to Table 5; these will upsell other products and increase average spent.
- **Targeted Retention Campaigns:** The 1,067 Lost Customers (Cluster 1) last purchased on average 248 days ago, so a re-engagement email campaign offering a personalized discount or new product highlight should be deployed to this segment before they are permanently lost. Certainly, given their historically low spend (avg. £481), a modest incentive is likely sufficient to prompt a return.
- **Loyalty Program for Potential Loyalists:** The 3,054 customers in Cluster 0 represent the largest segment and the greatest growth opportunity. With low average frequency (3.7 orders) but relatively recent engagement (43.7 days), these customers are active but not yet habitual, hence a loyalty program (like with points) or volume discount structure could convert this group into Loyal Customers over time, boosting revenue.
- **Wholesale account management:** The 13 Champions (Cluster 2) generate an average of £127,338 each and are almost certainly wholesale buyers. Consequently, these accounts should be managed individually with dedicated account managers, priority fulfilment, and bespoke pricing agreements, as losing even one represents a significant revenue risk to the business. Moreover, revenue figures reported at company level should be interpreted with care, as these 13 accounts alone significantly inflate aggregate metrics such as mean invoice value.

5.3 Limitations and Future Analysis

Evidently, this analysis is retrospective and based solely on transaction data, hence several limitations should be acknowledged. First, no demographic or behavioral data beyond purchasing history is available, which limits the depth of customer profiling possible. Second, the association rules reflect UK purchasing behavior only and may not generalize to international markets. Third, the dataset ends in December 2011, and findings may not reflect current consumer trends.

Furthermore, future work could incorporate text mining of product descriptions to cluster items by category, apply time-series forecasting to predict seasonal demand peaks, and extend the RFM model with additional dimensions such as product category diversity.

6. Conclusion

This project applied a full data mining pipeline to the UCI Online Retail II dataset, addressing three research questions defined in Section 1 from the perspective of a data scientist at a UK-based gift retailer.

The preprocessing pipeline reduced the raw dataset from 541,910 to 397,885 clean records, removing missing values, cancelled orders, and invalid entries. Then, the exploratory data analysis revealed a strongly seasonal revenue pattern peaking in November 2011, a UK-dominated customer base, and a right-skewed revenue distribution indicative of a small number of high-value transactions.

Thereafter, three data mining techniques were successfully applied. Association rule mining identified 41 rules using an Apriori algorithm, with the strongest rule achieving a lift of 19.37, confirming significant co-purchase patterns within the Regency tableware range. Additionally, outlier detection flagged 99 anomalous invoices accounting for 14.5% of total revenue, revealing the presence of wholesale buyers within the customer base. Finally, customer segmentation using RFM analysis and K-Means clustering produced four well-separated groups (silhouette score = 0.616), enabling targeted marketing strategies for each segment.

Collectively, these findings demonstrate that transaction data alone contains sufficient structure to generate actionable business intelligence. The four recommendations proposed in Section 5 (product bundling, retention campaigns, a loyalty program, and wholesale account management) address each identified segment and pattern directly, providing the retailer with a data-driven foundation for improving both customer retention and revenue growth.

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